Examination practice — Speed, energy, work and power

2A — 30 October 2026

Take $g = 10\ \text{N/kg}$ unless stated otherwise. Show all your working, write the equation in symbols before the numbers, and give units. The numbers in square brackets are the marks for each part.

A Speed and distance–time graphs

Exercise 1 — The round trip

A driver travels 60 km to a town at an average speed of 90 km/h and returns along the same road at an average speed of 60 km/h.

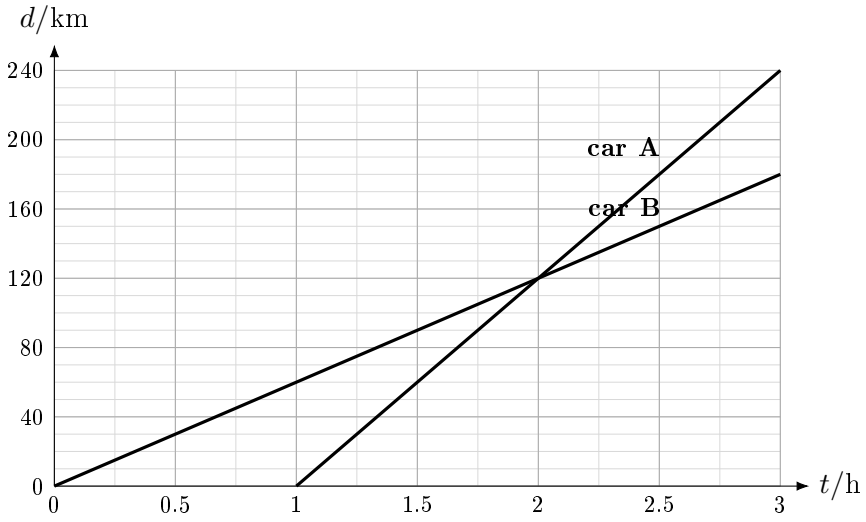
(a) A student says: "The average speed for the whole trip is $(90 + 60)/2 = 75$ km/h." Without calculating, say whether the true average speed is greater than, less than, or equal to 75 km/h. Justify in one line. **[2]**

(b) Calculate the time taken for each part of the trip, in hours, and the average speed for the whole trip. **[3]**

(c) Express the average speed of the whole trip in m/s. **[1]**

Exercise 2 — Overtaking

The graph shows the motion of two cars, A and B, on the same straight road. Both start from the same point P. Car B leaves first.



(a) Use the graph to calculate the speed of car A and the speed of car B. **[2]**

(b) State how the graph shows that car A overtakes car B, and read from the graph the time and the distance from P at which this happens. **[2]**

(c) Calculate how far ahead of car A is car B at $t = 1.5$ h. **[2]**

(d) Car A now leaves 0.5 h later than in the graph (at $t = 1.5$ h) but with the same speed. Without drawing, find the time at which A reaches B. Explain your method. **[2]**

Exercise 3 — Timing a trolley

A student measures the time taken by a trolley to travel 1.50 m down a slope. She repeats the measurement five times with a stopwatch.

run	1	2	3	4	5
time / s	3.0	2.9	3.1	4.3	3.0

(a) Identify the anomalous result and suggest one likely cause. **[2]**

(b) Calculate the mean time, leaving out the anomalous result. **[1]**

(c) Calculate the average speed of the trolley. **[2]**

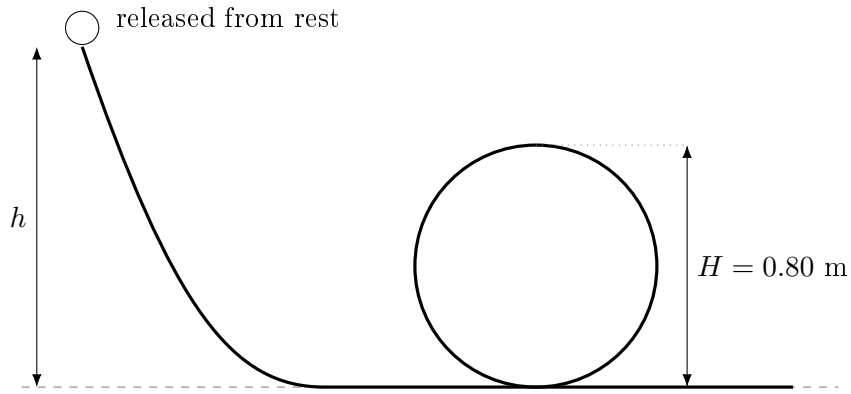
(d) The student always starts the stopwatch 0.2 s after releasing the trolley. State the type of error and say whether her calculated speed is too large or too small. Explain. **[2]**

(e) Suggest one change to the experiment that reduces the error in the time. **[1]**

B Conservation of energy

Exercise 4 — The loop

A trolley of mass 0.50 kg is released from rest on a frictionless track at height h above the lowest point of the track. The track has a vertical loop. The top of the loop is $H = 0.80$ m above the lowest point.



To stay on the track at the top of the loop, the trolley must be moving at **at least 2.0 m/s** there (this is given: you do not need to prove it). Take $g = 10$ N/kg.

(a) A student says: "Release the trolley from the same height as the top of the loop, 0.80 m." Without calculating, explain why the trolley will not stay on the track at the top. **[2]**

(b) Write, in symbols and without numbers, the equation that gives the **minimum** release height h in terms of H, the speed v at the top, and g. **[2]**

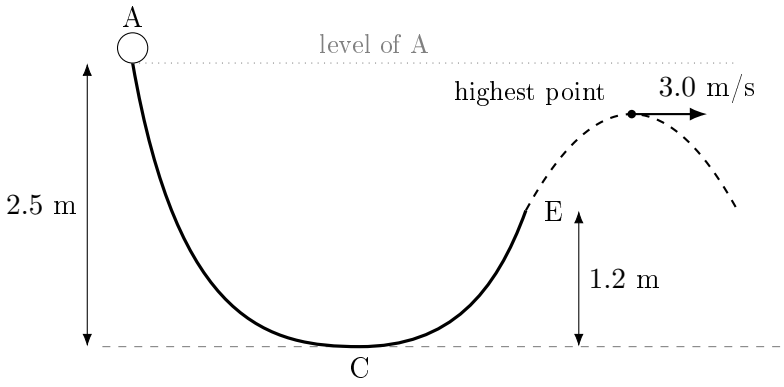
(c) Calculate the minimum release height. **[2]**

(d) The trolley is replaced by one of mass 1.0 kg. State the new minimum release height and explain. **[1]**

(e) The trolley (0.50 kg) is released from $h = 1.5$ m. Calculate its speed at the lowest point and at the top of the loop. Does it stay on the track? **[3]**

Exercise 5 — A skater leaves the half-pipe at an angle

A skater of mass 55 kg starts from rest at A, 2.5 m above the lowest point C of a half-pipe. She leaves the right-hand ramp at E, 1.2 m above C, moving **upwards at an angle**. At the highest point of her flight she is moving **horizontally** at 3.0 m/s. Ignore friction and air resistance. Take $g = 10$ N/kg.



(a) Without calculating, say whether her highest point is higher than A, lower than A, or at the same height as A. Explain in one line. **[2]**

(b) Calculate her speed at E. **[2]**

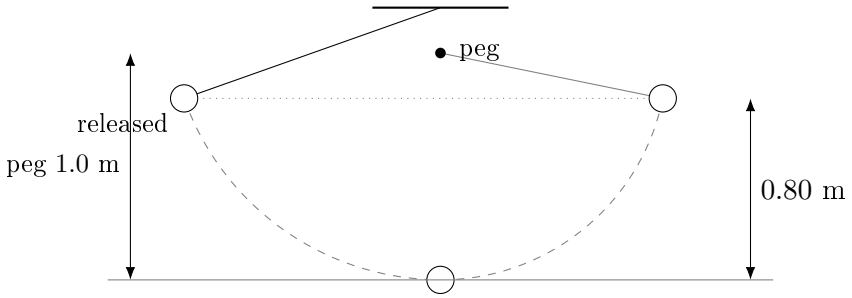
(c) Write, in symbols and without numbers, the equation that gives the height $h_{\max}$ of her highest point above C in terms of h_A, v and g. **[2]**

(d) Calculate $h_{\max}$. **[2]**

(e) A skater of mass 110 kg does the same, with the same horizontal speed 3.0 m/s at the top. State her highest point and explain. **[1]**

Exercise 6 — Pendulum with a peg

A pendulum bob of mass 0.20 kg hangs from a string 1.2 m long. It is released from rest with the string taut, when the bob is 0.80 m above its lowest point. A thin peg is fixed 1.0 m above the lowest point of the bob: when the string passes the vertical it catches on the peg and the bob continues on a shorter arc. Ignore friction and air resistance. Take $g = 10$ N/kg.



(a) Calculate the speed of the bob at its lowest point. **[2]**

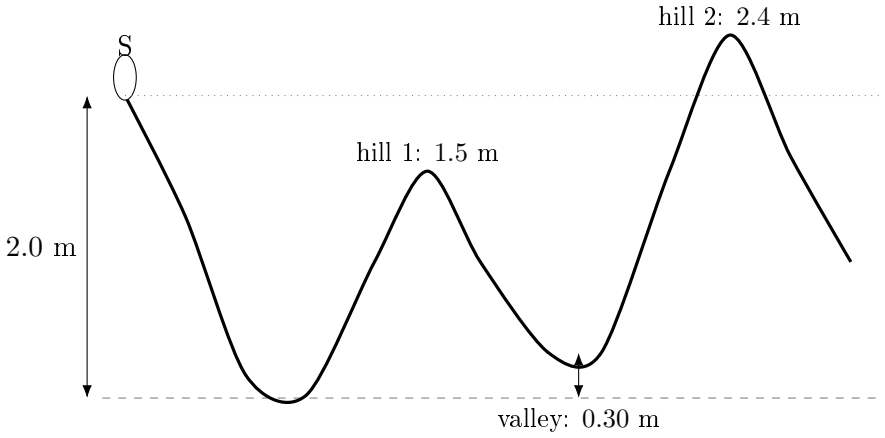
(b) State the height above the lowest point to which the bob rises on the peg side. Explain without calculating. **[2]**

(c) In a real experiment the bob rises slightly less than this. Give one reason. **[1]**

(d) The bob is replaced by one of mass 0.40 kg. State the effect on the speed at the lowest point. **[1]**

Exercise 7 — The track with two hills

A ball of mass 0.10 kg is released from rest at S on a smooth track. Heights are measured above the lowest level of the track. Ignore friction and air resistance. Take $g = 10$ N/kg.



(a) Without calculating, state which of the two hills the ball can pass over. Explain. **[2]**

(b) Calculate the speed of the ball at the top of hill 1. **[3]**

(c) Describe what happens when the ball reaches hill 2 and state the height at which it is momentarily at rest. **[2]**

(d) Calculate the speed of the ball in the valley that is 0.30 m above the lowest level. **[2]**

(e) The ball is replaced by one of mass 0.20 kg. State whether it can pass over hill 2 and explain. **[1]**

Exercise 8 — Speed and height: from data to g

A student lets a trolley roll from rest down a smooth slope. She measures its speed v after it has dropped through a vertical height h.

h / m	0.50	1.00	1.50	2.00
v / (m/s)	3.2	4.5	5.5	6.3
v^2 / $(\text{m/s})^2$				

(a) Complete the last row of the table. **[2]**

(b) She plots v^2 (vertical axis) against h (horizontal axis). Describe the shape of the graph she expects. **[1]**

(c) Starting from conservation of energy, write v^2 in terms of g and h and show that the gradient of the graph is $2g$. **[2]**

(d) Use the last data point to calculate the gradient and hence a value for g. **[2]**

(e) Predict the speed of the trolley after a drop of 8.0 m. **[2]**

Exercise 9 — Air resistance: energy bookkeeping

A ball of mass 2.0 kg is dropped from rest from a height of 5.0 m. Air resistance acts on the ball. The table shows the energy stores at three positions. Take $g = 10$ N/kg.

position	gravitational potential / J	kinetic / J	thermal energy transferred to the air / J
top ($h = 5.0$ m)		0	0
midway ($h = 2.5$ m)		40	
bottom ($h = 0$)		75	

(a) Complete the table. **[3]**

(b) Calculate the speed of the ball at the bottom. **[2]**

(c) State the speed the ball would have at the bottom without air resistance. **[1]**

(d) The energy transferred to the thermal store equals $F \times d$, where F is the force of air resistance and d the distance fallen. Calculate the average force of air resistance in the upper half and in the lower half of the fall. **[3]**

(e) Explain, in one line, why the two values are different. **[1]**

C Doing work

Exercise 10 — A crane lifts a load

A crane lifts a load of mass 500 kg through 12 m. Take $g = 10$ N/kg.

(a) In the first lift the crane pulls the load with a force of 5000 N. Calculate the work done by the crane and the gain in gravitational potential energy of the load. Compare the two values. **[3]**

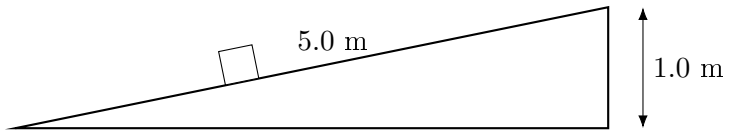
(b) In a second lift the crane pulls with a constant force of 5600 N, and the load moves faster and faster. Calculate the work done by the crane over the same 12 m. **[1]**

(c) The work done is greater than the gain in gravitational potential energy. In which energy store does the difference end up? **[1]**

(d) Calculate the speed of the load at a height of 12 m in the second lift. **[2]**

Exercise 11 — A ramp does not reduce the work

A box of mass 30 kg must be raised onto a platform 1.0 m high. In both cases below the box gains no kinetic energy and friction is ignored. Take $g = 10$ N/kg.



(a) Option 1: the box is lifted vertically by hand. Calculate the work done by the person. **[2]**

(b) Calculate the force the person needs to apply in option 1. **[1]**

(c) Option 2: the box is pushed up a smooth ramp 5.0 m long. State the work done by the person and explain. **[2]**

(d) Calculate the force along the ramp in option 2. **[2]**

(e) The ramp is replaced by one 10 m long (same platform). State the new force and the new work. **[2]**

(f) A student says: "A ramp reduces the work needed." Comment. **[1]**

Exercise 12 — Stopping distances

A car of mass 1000 kg travelling at 20 m/s brakes to a stop over a distance of 40 m. The braking force is constant.

(a) Calculate the kinetic energy of the car before braking. **[1]**

(b) Calculate the braking force. **[2]**

(c) For each change below, calculate the new stopping distance. Each change is to the original situation, and the braking force stays the same unless stated.

(i) the mass of the car is doubled (2000 kg), same speed **[1]**

(ii) the speed is doubled (40 m/s), same mass **[2]**

(iii) the braking force is doubled (10 000 N), same mass and speed **[1]**

(iv) the mass is halved (500 kg) **and** the speed is doubled (40 m/s) **[2]**

Exercise 13 — A ball in the sand

A ball of mass 0.50 kg is dropped from rest from a height of 3.0 m above sand. It sinks 4.0 cm into the sand before stopping. Ignore air resistance. Take $g = 10$ N/kg.

(a) Calculate the speed of the ball when it reaches the surface of the sand. **[2]**

(b) State the kinetic energy of the ball at the surface of the sand. **[1]**

(c) Calculate the total energy transferred to the sand and the ball while the ball stops. (Think: the ball also moves down 4.0 cm.) **[2]**

(d) Calculate the average force exerted by the sand on the ball. **[2]**

(e) With softer sand, the ball sinks 8.0 cm. Estimate the average force. Is it about double, about half, or the same as before? **[1]**

D Power and efficiency

Exercise 14 — Three cyclists

Three cyclists climb the same hill, which rises 120 m. Ignore the mass of the bikes. Take $g = 10$ N/kg.

cyclist	mass / kg	time to climb
Anna	60	10 min
Ben	90	15 min
Chen	60	8.0 min

(a) Calculate the work done against gravity by each cyclist. **[3]**

(b) Calculate the power of each cyclist. **[3]**

(c) Ben says: "I took longer than Anna, so my power was smaller." Is he right? Explain. **[2]**

(d) State who transferred the most energy and who had the greatest power. **[1]**

Exercise 15 — The chain of efficiencies

A power station burns fuel releasing 500 MJ of chemical energy. The power station is 40% efficient in producing electrical energy. The transmission cables are 90% efficient. An electric kettle is 80% efficient in transferring energy to the thermal energy of the water.

(a) Calculate the electrical energy produced by the power station. **[1]**

(b) Calculate the electrical energy that reaches the kettle. **[1]**

(c) Calculate the useful energy transferred to the water. **[1]**

(d) Calculate the overall efficiency of the whole chain. **[2]**

(e) A student says the overall efficiency is the mean of the three, 70%. Without calculating, explain why the overall efficiency must be less than 40%. **[2]**

(f) A gas burner that is 70% efficient heats the same water. Calculate the chemical energy of the gas it needs to transfer the same useful energy to the water, and compare it with 500 MJ. **[2]**

Exercise 16 — Two pumps

Two pumps lift water through a vertical height of 12 m. Pump A has a power input of 2.0 kW and is 60% efficient. Pump B has a power input of 1.5 kW and is 90% efficient. Take $g = 10$ N/kg.

(a) Calculate the useful power output of each pump. **[2]**

(b) Calculate the mass of water lifted by each pump in one minute. **[3]**

(c) Calculate the energy wasted by each pump in one minute. **[2]**

(d) A technician says: "Pump A has the greater power, so it lifts more water." Comment. **[1]**

E Energy resources

Exercise 17 — A hydroelectric power station

In a hydroelectric power station, 2.0×10^3 kg of water falls through a height of 50 m every second. The generator is 80% efficient. Take $g = 10$ N/kg.

(a) Calculate the power available from the falling water. **[2]**

(b) Calculate the electrical power output. **[1]**

(c) A town needs 4.0 MW. How many identical stations are needed? **[1]**

(d) In the dry season the height of the fall is only 30 m, with the same flow of water. Calculate the new electrical power output. **[2]**

(e) State one advantage and one disadvantage of hydroelectric power. **[2]**

Exercise 18 — Choosing energy resources

The table lists six energy resources.

resource	**renewable (R) or non-renewable (N)**
coal	
wind	
geothermal	
nuclear fission	
tidal	
biofuel	

(a) Complete the table. **[3]**

(b) Coal, wind and biofuel all depend on energy that came from the Sun. Name **two** of the other three resources and state the source of their energy. **[2]**

(c) A student says: "Nuclear fission is non-renewable because it releases radioactive waste." Explain what is wrong with this reason and give the correct one. **[2]**

(d) A country plans to generate all its electricity from solar cells and wind turbines. Explain one problem and suggest one way to reduce it. **[2]**

F Paper 2 style: multiple choice

Exercise 19 — Multiple choice

For each question choose the correct answer, A, B, C or D. Take $g = 10$ N/kg.

1. A skater leaves a ramp moving upwards at an angle. Which statement is correct about her energy at the highest point of the flight? **[1]**
   A Her kinetic energy is zero. **B** Her gravitational potential energy is zero.
   C She still has kinetic energy because she is still moving horizontally. **D** Her total energy is zero.

2. An object has its mass multiplied by 3 and its speed halved. Its kinetic energy is multiplied by: **[1]**
   A 0.75 **B** 1.5 **C** 6 **D** 12

3. A force of 20 N moves an object 5.0 m in 4.0 s. The power developed is: **[1]**
   A 100 W **B** 80 W **C** 25 W **D** 1.25 W

4. A machine takes in 800 J of energy and transfers 200 J usefully. Its efficiency is: **[1]**
   A 75% **B** 25% **C** 33% **D** 400%

5. Which is the reason why nuclear fission is a non-renewable energy resource? **[1]**
   A It produces radioactive waste. **B** It releases carbon dioxide.
   C It needs cooling water. **D** The supply of uranium is finite.

6. A distance–time graph is a straight line through the origin and through the point (4.0 s, 20 m). The speed at $t = 8.0$ s is: **[1]**
   A 2.5 m/s **B** 10 m/s **C** 5.0 m/s **D** 40 m/s

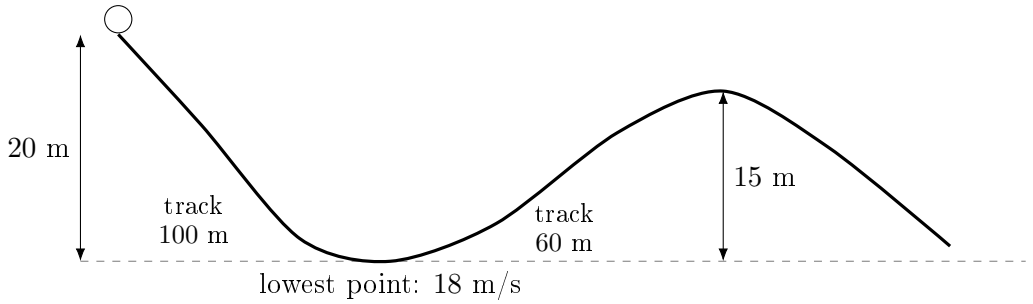
7. A person holds a 10 kg bag still for 60 s. The work done on the bag is: **[1]**
   A 6000 J **B** 600 J **C** 100 J **D** 0 J

8. A ball dropped from rest from a height H reaches the ground with speed v. To reach the ground with speed $2v$ it must be dropped from: **[1]**
   A $2H$ **B** $\sqrt{2}\,H$ **C** $4H$ **D** $8H$

G Challenge

Exercise 20 — The roller coaster with friction

A roller-coaster car of mass 400 kg is released from rest at the top of a hill, 20 m above the lowest point of the track. At the lowest point it is moving at 18 m/s. The track from the top to the lowest point is 100 m long. Take $g = 10$ N/kg.



(a) Calculate the kinetic energy of the car at the lowest point. **[2]**

(b) Calculate the gravitational potential energy lost by the car between the top and the lowest point. **[1]**

(c) State the energy transferred to the thermal store between the top and the lowest point. **[1]**

(d) Calculate the average friction force on the car. **[2]**

(e) After the lowest point there is a second hill, 15 m high, reached after 60 m of track. Assume the same average friction force. By calculating the energy needed, decide whether the car reaches the top of the second hill. **[3]**

(f) State whether the car would reach the top of the second hill if there were no friction. Justify. **[1]**